

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12397129
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Welch; Jeffrey M. et al.

Coronary guide catheter

Abstract

Embodiments include an interventional guide catheter, comprising: a main tubular shaft with a distal tip and proximal end; the main tubular shaft comprising: a main inner structural layer comprising a metallic helically wound multi-filar wire extending from a proximal tube termination to a distal end, a braided wire layer covering the main inner structural layer that extends from the proximal tube termination to the distal end, an outer layer of polymer jacketing fixedly attached to the main inner structural layer and braid layer, an inner layer of polymer jacketing and fixedly attached to the main inner structural layer, a distal tip made of layers of polymer, a distal end curve shape for anatomical conformance that is heat processed in the main metal portion of the structure; and a lamination of the inner layer, main inner structural layer, braided wire layer and outer layer. Other embodiments are also included herein.

Inventors: Welch; Jeffrey M. (Maple Grove, MN), Sutton; Gregg Stuart (Maple Grove, MN)

Applicant: NorMedix, Inc. (Eden Prairie, MN)

Family ID: 1000008779411

Assignee: NorMedix, Inc. (Eden Prairie, MN)

Appl. No.: 17/668725

Filed: February 10, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220233810 A1	Jul. 28, 2022

Related U.S. Application Data

continuation parent-doc US 15224323 20160729 ABANDONED child-doc US 17668725
us-provisional-application US 62199050 20150730

Publication Classification

Int. Cl.: A61M25/00 (20060101); A61M25/06 (20060101)

U.S. Cl.:

CPC A61M25/005 (20130101); A61M25/0045 (20130101); A61M25/0041 (20130101); A61M25/0662 (20130101)

Field of Classification Search

CPC: A61M (25/005); A61M (25/0041); A61M (25/0045); A61M (25/0662); A61M (2025/0681); A61M (2025/0047)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4493329	12/1984	Crawford et al.	N/A	N/A
4737153	12/1987	Shimamura	138/131	A61M 16/0425
5106363	12/1991	Nobuyoshi	N/A	N/A
5472435	12/1994	Sutton	N/A	N/A
5558652	12/1995	Henke	N/A	N/A
5573522	12/1995	Houser et al.	N/A	N/A
5658264	12/1996	Samson	N/A	N/A
5695483	12/1996	Samson	N/A	N/A
5755704	12/1997	Lunn	N/A	N/A
5795341	12/1997	Samson	N/A	N/A
5846229	12/1997	Berg	604/528	A61M 25/0041
5853400	12/1997	Samson	N/A	N/A
5876386	12/1998	Samson	N/A	N/A
5891112	12/1998	Samson	N/A	N/A
5891114	12/1998	Chien et al.	N/A	N/A
5947940	12/1998	Beisel	604/526	A61M 25/005
5951929	12/1998	Wilson et al.	N/A	N/A
5957910	12/1998	Holden, II et al.	N/A	N/A
6042578	12/1999	Dinh et al.	N/A	N/A
6053903	12/1999	Samson	N/A	N/A
6053904	12/1999	Scribner et al.	N/A	N/A
6103037	12/1999	Wilson	N/A	N/A
6143013	12/1999	Samson et al.	N/A	N/A
6152911	12/1999	Giannoble	N/A	N/A
6165163	12/1999	Chien et al.	N/A	N/A
6217565	12/2000	Cohen	N/A	N/A
6355027	12/2001	Le et al.	N/A	N/A
6368316	12/2001	Jansen et al.	N/A	N/A

6374476	12/2001	Ponzi et al.	N/A	N/A
6451005	12/2001	Saitou	N/A	N/A
6464684	12/2001	Galdonik	N/A	N/A
6562022	12/2002	Hoste et al.	N/A	N/A
6589227	12/2002	Sonderskov	N/A	N/A
6616651	12/2002	Stevens	N/A	N/A
6626889	12/2002	Simpson et al.	N/A	N/A
6629952	12/2002	Chien et al.	N/A	N/A
6635047	12/2002	Forsberg	N/A	N/A
6689120	12/2003	Gerdts	604/526	A61M 25/005
6702972	12/2003	Markle	N/A	N/A
6709429	12/2003	Schaefer et al.	N/A	N/A
6740073	12/2003	Saville	N/A	N/A
6824553	12/2003	Samson et al.	N/A	N/A
6866660	12/2004	Garabedian et al.	N/A	N/A
6942654	12/2004	Schaefer et al.	N/A	N/A
7018372	12/2005	Casey et al.	N/A	N/A
7025758	12/2005	Klint	N/A	N/A
7104979	12/2005	Jansen et al.	N/A	N/A
7117703	12/2005	Kato et al.	N/A	N/A
7331948	12/2007	Skarda	N/A	N/A
7354428	12/2007	Gosiengfiao et al.	N/A	N/A
7455739	12/2007	Zhou	N/A	N/A
7507229	12/2008	Hewitt et al.	N/A	N/A
7534317	12/2008	Brustad et al.	N/A	N/A
7597830	12/2008	Zhou	N/A	N/A
7615043	12/2008	Zhou	N/A	N/A
7637902	12/2008	Eversull et al.	N/A	N/A
7674239	12/2009	Sisken et al.	N/A	N/A
7704245	12/2009	Dittman et al.	N/A	N/A
7713259	12/2009	Gosiengfiao et al.	N/A	N/A
7744587	12/2009	Murphy	N/A	N/A
7763012	12/2009	Petrick et al.	N/A	N/A
7815762	12/2009	Lentz et al.	N/A	N/A
7824392	12/2009	Zhou	N/A	N/A
7833218	12/2009	Lunn et al.	N/A	N/A
7850623	12/2009	Griffin et al.	N/A	N/A
7909779	12/2010	Shimogami et al.	N/A	N/A
7909812	12/2010	Jansen	N/A	N/A
7914515	12/2010	Heideman et al.	N/A	N/A
7955313	12/2010	Boismier	N/A	N/A
7968038	12/2010	Dittman et al.	N/A	N/A
8021352	12/2010	Slazas et al.	N/A	N/A
8034045	12/2010	Lyons	N/A	N/A
8070898	12/2010	Eversull et al.	N/A	N/A
8100881	12/2011	Hoffa	N/A	N/A
8114144	12/2011	Chow et al.	N/A	N/A
8142415	12/2011	Warnock, Jr. et al.	N/A	N/A
8157790	12/2011	Kubo	N/A	N/A

8182466	12/2011	Stehr et al.	N/A	N/A
8206373	12/2011	Zhou	N/A	N/A
8251976	12/2011	Zhou	N/A	N/A
8257314	12/2011	Agnew	N/A	N/A
8317772	12/2011	Jansen et al.	N/A	N/A
8343136	12/2012	Howat et al.	N/A	N/A
8348925	12/2012	Fischell et al.	N/A	N/A
8366699	12/2012	Jimenez	N/A	N/A
8377035	12/2012	Zhou et al.	N/A	N/A
8431057	12/2012	Guo et al.	N/A	N/A
8475431	12/2012	Howat	N/A	N/A
8486048	12/2012	Kubo et al.	N/A	N/A
8529719	12/2012	Pingleton et al.	N/A	N/A
8535293	12/2012	Faherty et al.	N/A	N/A
8535294	12/2012	Fischell et al.	N/A	N/A
8540695	12/2012	Shimogami et al.	N/A	N/A
8591495	12/2012	Fischell et al.	N/A	N/A
8636270	12/2013	Ostrovsky	N/A	N/A
8647323	12/2013	Guo	604/527	A61M 25/0043
8696699	12/2013	Chomas et al.	N/A	N/A
8702679	12/2013	Deckman et al.	N/A	N/A
8702680	12/2013	Jimenez et al.	N/A	N/A
8708997	12/2013	Parker	N/A	N/A
8721625	12/2013	Klint	N/A	N/A
8758326	12/2013	Hennissy	N/A	N/A
8858530	12/2013	Nishigishi et al.	N/A	N/A
8864744	12/2013	Howat et al.	N/A	N/A
8926560	12/2014	Dinh et al.	N/A	N/A
8986284	12/2014	Kuwada et al.	N/A	N/A
9114229	12/2014	Fuentes	N/A	N/A
9254374	12/2015	Thorstenson et al.	N/A	N/A
9259550	12/2015	Schaeffer	N/A	N/A
9259813	12/2015	Heideman	N/A	N/A
9278191	12/2015	Nihonmatsu et al.	N/A	N/A
2002/0068899	12/2001	James, Jr. et al.	N/A	N/A
2002/0095102	12/2001	Winters	N/A	N/A
2003/0009184	12/2002	Pepin	N/A	N/A
2003/0191451	12/2002	Gilmartin	N/A	N/A
2004/0092844	12/2003	Johnson et al.	N/A	N/A
2005/0043713	12/2004	Zhou	N/A	N/A
2005/0090802	12/2004	Connors, III et al.	N/A	N/A
2006/0095050	12/2005	Hartley	606/108	A61M 25/0662
2006/0111649	12/2005	Zhou	N/A	N/A
2007/0233040	12/2006	Macnamara et al.	N/A	N/A
2010/0217235	12/2009	Thorstenson et al.	N/A	N/A
2012/0109078	12/2011	Schaeffer	N/A	N/A
2012/0130217	12/2011	Kauphusman et al.	N/A	N/A
2012/0172798	12/2011	Miller et al.	N/A	N/A

2012/0277729	12/2011	Melsheimer	N/A	N/A
2013/0018318	12/2012	Ravichandran et al.	N/A	N/A
2013/0197481	12/2012	Guo et al.	N/A	N/A
2013/0255062	12/2012	Howat	N/A	N/A
2013/0289697	12/2012	Baker et al.	N/A	N/A
2014/0046138	12/2013	Merk et al.	N/A	N/A
2014/0214006	12/2013	Hiroshige et al.	N/A	N/A
2014/0236124	12/2013	Miyata	N/A	N/A
2014/0330253	12/2013	Weiss	N/A	N/A
2014/0352871	12/2013	Wang et al.	N/A	N/A
2015/0051541	12/2014	Kanemasa et al.	N/A	N/A
2015/0119859	12/2014	Cajamarca et al.	N/A	N/A
2015/0174363	12/2014	Sutermeister et al.	N/A	N/A
2015/0217083	12/2014	Richter et al.	N/A	N/A
2015/0231360	12/2014	Watanabe et al.	N/A	N/A
2015/0258306	12/2014	Plassman et al.	N/A	N/A
2015/0335857	12/2014	Ishikawa	N/A	N/A
2016/0001040	12/2015	Yamaguchi et al.	N/A	N/A
2016/0121075	12/2015	Schaeffer	N/A	N/A
2017/0028167	12/2016	Welch et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2994262	12/2022	CA	N/A
0810003	12/1996	EP	N/A
2687254	12/2013	EP	N/A
3328476	12/2018	EP	N/A
H09506541	12/1996	JP	N/A
2005537877	12/2004	JP	N/A
2018522704	12/2017	JP	N/A
WO-0166176	12/2000	WO	N/A
WO-2006031874	12/2005	WO	N/A
WO-2014210427	12/2013	WO	N/A
WO-2017020012	12/2016	WO	N/A

OTHER PUBLICATIONS

“U.S. Appl. No. 15/224,323, Appeal Brief filed Feb. 15, 2021”, 22 pgs. cited by applicant

“U.S. Appl. No. 15/224,323, Appeal Decision mailed Dec. 10, 2021”, 9 pgs. cited by applicant

“U.S. Appl. No. 15/224,323, Examiners Answer mailed Mar. 19, 2021”, 7 pgs. cited by applicant

“U.S. Appl. No. 15/224,323, Final Office Action mailed Jun. 7, 2019”, 13 pgs. cited by applicant

“U.S. Appl. No. 15/224,323, Final Office Action mailed Jul. 14, 2020”, 14 pgs. cited by applicant

“U.S. Appl. No. 15/224,323, Non Final Office Action mailed Jan. 8, 2020”, 14 pgs. cited by applicant

“U.S. Appl. No. 15/224,323, Non Final Office Action mailed Mar. 26, 2018”, 14 pgs. cited by applicant

“U.S. Appl. No. 15/224,323, Non Final Office Action mailed Nov. 23, 2018”, 13 pgs. cited by applicant

“U.S. Appl. No. 15/224,323, Reply Brief mailed May 19, 2022”, 9 pgs. cited by applicant

“U.S. Appl. No. 15/224,323, Response filed Jun. 8, 2020 to Non Final Office Action mailed Jan. 8, 2020”, 14 pgs. cited by applicant
“U.S. Appl. No. 15/224,323, Response filed Dec. 9, 2019 to Final Office Action mailed Jun. 7, 2019”, 15 pgs. cited by applicant
“U.S. Appl. No. 15/224,323, Response to Non Final Office Action mailed Mar. 26, 2018 filed Sep. 26, 2018”, 13 pgs. cited by applicant
“U.S. Appl. No. 15/224,323, Response to Non Final Office Action mailed Nov. 23, 2018 filed Mar. 25, 2019”, 11 pgs. cited by applicant
“Canadian Application Serial No. 2,994,262, Examiner's Rule 86(2) Requisition mailed Aug. 24, 2022”, 6 pgs. cited by applicant
“Canadian Application Serial No. 2,994,262, Response filed Dec. 23, 2022 to Examiner's Rule 86(2) Requisition mailed Aug. 24, 2022”, 38 pgs. cited by applicant
“European Application Serial No. 16748030.0, Communication Pursuant to Article 94(3) EPC mailed Nov. 28, 2018”, 7 pgs. cited by applicant
“European Application Serial No. 16748030.0, Response filed Apr. 5, 2019 to Communication Pursuant to Article 94(3) EPC mailed Nov. 28, 2018”, 35 pgs. cited by applicant
“European Application Serial No. 16748030.0, Response filed Oct. 1, 2018 to Communication Pursuant to Rules 161(1) and 162 EPC mailed Mar. 20, 2018”, 22 pgs. cited by applicant
“International Application Serial No. PCT/US2016/044881, International Preliminary Report on Patentability mailed Feb. 8, 2018”, 9 pgs. cited by applicant
“International Application Serial No. PCT/US2016/044881, International Search Report mailed Nov. 18, 2016”, 6 pgs. cited by applicant
“International Application Serial No. PCT/US2016/044881, Written Opinion mailed Nov. 18, 2016”, 7 pgs. cited by applicant
“Japanese Application Serial No. 2018-525532, Notification of Reasons for Refusal mailed Jul. 7, 2020”, with English translation, 17 pages. cited by applicant
“Japanese Application Serial No. 2018-525532, Office Action mailed Feb. 9, 2021”, with English translation, 14 pages. cited by applicant
“Japanese Application Serial No. 2018-525532, Response filed Jan. 6, 2021 to Notification of Reasons for Refusal mailed Jul. 7, 2020”, with English claims, 7 pages. cited by applicant
“Japanese Application Serial No. 2018-525532, Response filed Aug. 6, 2021 to Office Action mailed Feb. 9, 2021”, w/ English claims, 7 pgs. cited by applicant

Primary Examiner: Osinski; Bradley J

Attorney, Agent or Firm: Schwegman Lundberg & Woessner, P.A.

Background/Summary

CLAIM OF PRIORITY (1) This application is a continuation of U.S. patent application Ser. No. 15/224,323, filed Jul. 29, 2016, which application claims the benefit of priority to U.S. Provisional Patent Application Ser. No. 62/199,050, filed Jul. 30, 2015, the contents of which are hereby incorporated by reference in their entireties.

FIELD OF THE TECHNOLOGY

(1) The present application relates to coronary guide catheters. More specifically, the present application relates to guide catheters that are used to introduce therapeutic catheters, such as stent delivery systems.

BACKGROUND

(2) Interventional guide catheters are used by physicians to place catheters, electrode leads and other therapeutic interventional devices to desired locations in the patient's body. The guide catheter provides support for device advancement (stents, balloons, etc.) for instance, to the coronary arteries. It is the main conduit for therapeutic device and guide wire transport and provides a means for injecting contrast media. Guide catheters typically have a pre-shaped distal end that is configured to allow access and direction into the arterial branches. In coronary guide catheters, the distal end shapes also provide back-up support for device placement by using shapes made to "back-up" against aortic anatomy.

(3) Guide catheters are made so that the distal shapes as much as possible, retain their shape during the procedure and don't soften in body temperature to any significant or detrimental degree. Guide catheters with distal end shapes also are typically made to provide end-to-end torque to allow "steering" the distal end into the artery. In addition, guide catheters are typically made to be relatively stiff so that there is no stretching and biasing during device passage. To achieve these performance criteria, current guide catheter construction means are usually polymer based with metallic wire braid reinforcement.

(4) The problems associated with current interventional guide catheters include softening at body temperature thereby losing critical performance features like back-up support or torsion. In addition, guide catheters have been prone to kinking, buckling, ovaling and stretching, especially in long procedures or difficult, more tortuous anatomy.

SUMMARY

(5) Embodiments disclosed herein include an interventional guide catheter for introducing interventional catheters into the vasculature, comprising: a main tubular shaft with a distal tip and proximal end; the main tubular shaft comprising: a main inner structural layer comprising a metallic helically wound multi-filar wire (wall thickness (0.0015-0.010") extending from a proximal tube termination to the distal end, a braided wire layer (0.0005-0.010" thick) covering the metallic helically wound multi-filar layer that extends from the proximal tube termination to the distal end, an outer layer of polymer jacketing covering and fixedly attached to the main metal structure and braid layer with wall thickness of 0.001-0.005", an inner layer of polymer jacketing covering and fixedly attached to the inner metal structure with wall thickness of 0.001-0.0005", a distal tip made of layers of polymer, the distal tip being 0.05-0.20" in length, an optional distal end curve shape for anatomical conformance that is heat processed in the main metal portion (e.g., one or more of the multi-filar layer, braid or the like) of the structure; and a lamination of the inner layer, metallic helically wound multi-filar layer, braid and outer layer that optionally does not comprise fusion of the outer and inner layers. The main metal portion is separately heat processed (e.g., to provide the distal end curve shape) prior to lamination in one example. In another example, the main metal portion is heat processed while incorporated with the other components of the catheter (e.g., the inner and outer layers, or the like).

(6) In an embodiment, the main helically wound multi-filar layer terminates distally before the primary curve of the distal end. In another embodiment, the main helically wound layer terminates proximal to the distal end curve shape including the primary curve.

(7) In an embodiment, the outer and inner layers are fused together through the main coil structure and braid.

(8) In an embodiment, the metal helically wound multi-filar layer comprises stainless steel.

(9) In an embodiment, the braid layer comprises stainless steel wire.

(10) In an embodiment, the multi-filar structure comprises at least 6 filars and not more than 20 filars.

(11) In an embodiment, the multi-filar structure wire has been swaged.

(12) In an embodiment, the outer diameter of the main tubular shaft is at least 0.060 inches and not more than 0.115 inches.

(13) In an embodiment, the outer layer comprises Pebax.

- (14) In an embodiment, the outer layer comprises nylon.
 - (15) In an embodiment, the outer layer is coated with a hydrophilic polymer.
 - (16) In an embodiment, the inner layer comprises PTFE.
 - (17) In an embodiment, the inner layer comprises nylon.
 - (18) In an embodiment, the inner layer is coated with a hydrophilic polymer.
 - (19) In an embodiment, the distal tip comprises a PTFE inner layer and a Pebax outer layer.
 - (20) In an embodiment, the metallic helically wound layer comprises welded terminations.
 - (21) In an embodiment, the metallic helically wound layer comprises a distal end that comprises a gold coating.
 - (22) In an embodiment, the gold coating is at least 0.5 mm and not more than 2 mm in length.
 - (23) In an embodiment, the outer layer comprises at least two layers of Pebax.
 - (24) In an embodiment, the outer layer is heat shrinkable to allow it to be formed tightly onto the metallic helically wound layer.
 - (25) In an embodiment, the metallic helically wound wire has a rectangular cross-section.
 - (26) In an embodiment, the metallic helically wound wire has a circular cross-section.
 - (27) In an embodiment, the metallic helically wound layer wire has an oval or elliptical cross-section.
 - (28) In an embodiment, the metallic helically wound layer wire has been coated with PTFE coating prior to forming into the multi-filar configuration.
 - (29) In an embodiment, the main tubular shaft length is at least 60 cm and not more than 200 cm long.
 - (30) This summary is an overview of some of the teachings of the present application and is not intended to be an exclusive or exhaustive treatment of the present subject matter. Further details are found in the detailed description and appended claims. Other aspects will be apparent to persons skilled in the art upon reading and understanding the following detailed description and viewing the drawings that form a part thereof, each of which is not to be taken in a limiting sense. The scope of the present application is defined by the appended claims and their legal equivalents.
-

Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) The technology may be more completely understood in connection with the following drawings, in which:
- (2) FIG. 1 is a front view of a guide catheter, according to an embodiment.
- (3) FIG. 2 is a back view of a guide catheter, according to an embodiment.
- (4) FIG. 3 is a cross-section view of a guide catheter, according to an embodiment.
- (5) FIG. 4 is a cross-section view of a portion of a guide catheter, according to an embodiment.
- (6) FIG. 5 is a cross-section view of a portion of a guide catheter, according to an embodiment.
- (7) FIG. 6 is a cross-section view of a portion of a guide catheter, according to an embodiment.
- (8) FIG. 7 is a cross-section view of a portion of a guide catheter, according to an embodiment.
- (9) FIG. 8 is a cross-section view of a portion of a guide catheter, according to an embodiment.
- (10) FIG. 9 is a cross-section view of a portion of a guide catheter, according to an embodiment.
- (11) FIG. 10 is a front view of a guide catheter, according to an embodiment.
- (12) While the technology is susceptible to various modifications and alternative forms, specifics thereof have been shown by way of example and drawings, and will be described in detail. It should be understood, however, that the application is not limited to the particular embodiments described. On the contrary, the application is to cover modifications, equivalents, and alternatives falling within the spirit and scope of the technology.

DETAILED DESCRIPTION

(13) The embodiments of the present technology described herein are not intended to be exhaustive or to limit the technology to the precise forms disclosed in the following detailed description. Rather, the embodiments are chosen and described so that others skilled in the art can appreciate and understand the principles and practices of the present technology.

(14) All publications and patents mentioned herein are hereby incorporated by reference. The publications and patents disclosed herein are provided solely for their disclosure. Nothing herein is to be construed as an admission that the inventors are not entitled to antedate any publication and/or patent, including any publication and/or patent cited herein.

(15) The guide catheter as described herein can solve the problems associated with current guide catheter technology by providing a novel design, construction and materials.

(16) The guide catheter, described herein, can be used in interventional cases where significant arterial tortuosity is encountered such as using a radial artery access or using a femoral approach on an obese patient.

(17) In various embodiments, the guide catheter can include a composite built tube that can be fabricated using a specially wound metal inner layer and jacketed with very thin layers of polymer inside and out. The metallic inner layer can be made using a multi-filar (6-20 filars) helically wound wire structure. In some embodiments, the helical structure can be swaged, such that each individual wire strand is partially rectangular in cross-section and therefore can result in a very tight/close fitting wire matrix. The helical structure can also be made using a non-swaged, round, square or rectangular wire.

(18) In various embodiments, the wall thickness of the inner metal structure can range from 1.5 to 10 thousandths of an inch thick.

(19) The helically wound metal structure can improve the mechanical integrity of the catheter tube, such as compared to current guide catheters with respect to kinking, buckling, flexibility, radial strength, and maintaining circularity of the catheter lumen cross-section.

(20) This marked improvement can be achieved by the significant increase in the amount of metal in the catheter. For instance, current guide catheters that are composite built or wire braid reinforced have total cross-sectional metallic component in the range of 5-10%. The guide catheter as described herein can have a total cross-sectional metallic component of 40-60%. The transmission of mechanical energy through this significantly higher modulus composite can result in significantly higher performance.

(21) The guide catheter of this invention also comprises an outer polymer layer and an inner polymer layer. In an embodiment, the outer polymer layer and the inner polymer layer can include one or more polymers, such as PTFE, Pebax, or Polyurethane. The polymer layers can be attached to the metal structure by thermal polymer heat-shrinking or reflow. The resultant wall thickness of the polymer layers can range from 0.5 to 3.0 thousandths of an inch for each layer.

(22) In various embodiments, the guide catheter can include a pre-shaped curve, such as a curved distal end region. The guide catheter can attain the pre-shaped curve configuration by heat-setting the metal in this portion of the catheter. The result can include a curve that retains its shape in body temperature and over time does not substantially soften, such as soften enough to unintentionally change shape.

(23) The guide catheter can further include a soft (low durometer) polymer distal tip, various distal curve shapes, a radiopaque distal marker band, and a proximal luer adapter. The guide catheter range in sizes from 4F to 8F and in lengths from 80 to 125 cm.

(24) In reference now to the figures, FIG. 1 shows a front view of a guide catheter **100**, according to an embodiment. FIG. 2 shows a back view of the guide catheter **100**. FIG. 3 shows a cross-sectional view of the guide catheter **100**. In an embodiment, the guide catheter **100** can be configured for introducing interventional catheters into the vasculature of a patient.

(25) In an embodiment, the catheter **100** can include a main tubular shaft **102** with a distal tip **104** and proximal end **106**. The distal tip **104** can be on the opposite end of the tubular shaft **102** from

the proximal end **106**. The distal tip **104** can include at least one layer of polymer. In an embodiment, the distal tip **104** includes at least two layers of polymer. In an embodiment, the distal tip **104** can include an inner layer and an outer layer. In an embodiment, the inner layer of the distal tip **104** can include PTFE. In an embodiment, the outer layer of the distal tip **104** can include Pebax.

(26) In an embodiment, the distal tip **104** can be at least 0.05 inches long. In an embodiment, the distal tip **104** can be at least 0.02 inches long. In an embodiment, the distal tip **104** can be 0.2 inches long or shorter. In an embodiment, the distal tip **104** can be 0.5 inches long or shorter. In various embodiments, the tubular shaft **102** can include a main inner structural layer.

(27) The main inner structural layer can include a metallic helically wound multi-filar wire extending from a proximal tube termination (e.g., the proximal shaft end **106** or proximal shaft portion **108**) to a distal end **112** including at least the bracketed **112** shown in FIGS. 1-3 (e.g., at the distal location). The main inner structural layer can further include a braided wire layer. In various embodiments, the braided wire layer can cover at least a portion of the outer portion (opposite from the inner lumen) of the metallic helically wound multi-filar layer that extends from the proximal tube termination to the distal end. In other embodiments, the braided wire layer is within the metallic helically wound multi-filar layer.

(28) In various embodiments, the main tubular shaft **102** can include an outer layer. The outer layer can include a polymer. The outer layer can jacket, coat, or cover the outer surface of the main inner structural layer. The outer layer can be fixedly attached to the main inner structural layer.

(29) In various embodiments, the main tubular shaft **102** can include an inner layer. The inner layer can include a polymer. The inner layer can jacket, coat, or cover the inner surface of the main inner structural layer. The inner layer can be fixedly attached to the main inner structural layer.

(30) The main tubular shaft **102** can include a curve, such as on the distal end (shown in FIG. 10). The curve shape can be configured for anatomical conformance. The shape can be heat processed in the main tubular shaft **102**, such as in the main inner structural layer or another metal portion. In an embodiment, the helically wound multi-filar layer terminates distally prior to the curve of the distal end. In another embodiment the helically wound multi-filar layer terminates proximal a primary curve of the curve (e.g., of the distal end curve shape) and distal to other portions of the curve including, but not limited to secondary and tertiary curves.

(31) In various embodiments, the metallic helically wound multi-filar layer, and the braid can be laminated by the inner layer and the outer layer, such that the lamination does not fuse the outer layer and the inner layer together.

(32) In an embodiment, the main tubular shaft **102** can be at least 60 cm long and not longer than 200 cm. In an embodiment, the main tubular shaft **102** can be at least 10 cm long and not longer than 300 cm. In an embodiment, the main tubular shaft **102** can be at least 30 cm long and not longer than 250 cm. In an embodiment, the main tubular shaft **102** can be at least 50 cm long and not longer than 225 cm.

(33) In an embodiment, the main tubular shaft **102** can have an outer diameter of at least 0.060 inches and not more than 0.115 inches. In an embodiment, the main tubular shaft **102** can have an outer diameter of at least 0.060 inches. In an embodiment, the main tubular shaft **102** can have an outer diameter of at least 0.040 inches. In an embodiment, the main tubular shaft **102** can have an outer diameter of at least 0.050 inches. In an embodiment, the main tubular shaft **102** can have an outer diameter of at least 0.070 inches. In an embodiment, the main tubular shaft **102** can have an outer diameter of at least 0.080 inches.

(34) In an embodiment, the main tubular shaft **102** can have an outer diameter of no greater than 0.115 inches. In an embodiment, the main tubular shaft **102** can have an outer diameter of no greater than 0.095 inches. In an embodiment, the main tubular shaft **102** can have an outer diameter of no greater than 0.105 inches. In an embodiment, the main tubular shaft **102** can have an outer diameter of no greater than 0.125 inches. In an embodiment, the main tubular shaft **102** can have an

outer diameter of no greater than 0.135 inches.

(35) FIG. 4 and FIG. 5 show cross-section views of portions of a guide catheter **100**, according to various embodiments. FIG. 5 shows a cross-section of a portion of the distal tip **104**. As seen in FIG. 5, the guide catheter **100** can define one or more apertures **506**. In various embodiments, the main tubular shaft **102** can define an aperture **506**. In an embodiment, the distal tip **104** can define an aperture **506**.

(36) FIG. 6 shows a cross-section view of a portion of the main tubular shaft **102**, according to an embodiment. FIG. 7 shows a cross-section view from the end of the main tubular shaft **102**. In an embodiment, the main tubular shaft **102**, can include a main inner structural layer **608**. The main inner structural layer **608** can include a metallic helically wound multi-filar wire. In various embodiments, the metallic helically wound multi-filar wire can include stainless steel. In various embodiments, the metallic helically wound multi-filar wire can be swaged.

(37) In various embodiments, the metallic helically wound multi-filar wire can include at least 6 filars and not more than 20 filars. In various embodiments, the metallic helically wound multi-filar wire can include at least 4 filars and not more than 24 filars. In various embodiments, the metallic helically wound multi-filar wire can include at least 8 filars and not more than 18 filars. In various embodiments, the metallic helically wound wire can have a rectangular cross-section, a circular cross-section, an oval cross-section or an elliptical cross-section. In various embodiments, the metallic helically wound wire can have been substantially coated with PTFE coating prior to forming into the multi-filar configuration.

(38) In an embodiment, the main inner structural layer **608** can include welded terminations. In an embodiment, the main inner structural layer **608** can include a distal end that includes a gold coating. In various embodiments, the gold coating can range from 0.5 mm thick to 2 mm in length. In various embodiments, the gold coating can range from 0.4 mm thick to 2.5 mm in length. In various embodiments, the gold coating can range from 0.25 mm thick to 3 mm in length.

(39) In an embodiment, the main inner structural layer **608** can have a thickness that can range from 0.0015 inches to 0.010 inches (e.g., one or more of a consistent thickness or variable thicknesses). In an embodiment, the main inner structural layer **608** can have a thickness of at least 0.0010 inches. In an embodiment, the main inner structural layer **608** can have a thickness of at least 0.0005 inches. In an embodiment, the main inner structural layer **608** can have a thickness of no greater than 0.015 inches. In an embodiment, the main inner structural layer **608** can have a thickness of no greater than 0.020 inches. Optionally, the main inner structural layer **608** includes a varying wall thickness. For instance, the metallic helically wound multi-filar wire is ground so that portions of the layer have varying thickness. The catheter, in some examples includes corresponding reduced diameter based on the grinding of the metallic helically wound multi-filar wire. In still another example, the metallic helically wound multi-filar wire is formed with a varied diameter (e.g., is necked) to accordingly decrease the thickness of the main inner structural layer **608**. In one example, a proximal portion of the main inner structural layer **608** includes a greater thickness relative to a distal portion to enhance pushability of the catheter. In another example, the distal portion of the main inner structural layer **608** has a lesser thickness than the proximal portion to facilitate bending and corresponding navigation through tortuous vasculature. In an embodiment, the main tubular shaft **102** can include an outer layer **610**. The outer layer **610** can include a polymer. The outer layer **610** can jacket or cover at least a portion of the outer portion of the main inner structural layer **608**. In an embodiment, the outer layer **610** can be at least 0.001 inches thick and not more than 0.005 inches thick. In an embodiment, the outer layer **610** can be at least 0.0007 inches thick. In an embodiment, the outer layer **610** can be at least 0.0005 inches thick. In an embodiment, the outer layer **610** can be no more than 0.007 inches thick. In an embodiment, the outer layer can be no more than 0.01 inches thick.

(40) In an embodiment, the outer layer **610** can include Pebax. In an embodiment, the outer layer **610** can include nylon. In an embodiment, the outer layer **610** can be coated with a hydrophilic

polymer. In an embodiment, the outer layer **610** can include at least two layers. In an embodiment, each of the two layers included in the outer layer **610** can include Pebax. In various embodiments, the outer layer **610** can be heat shrinkable, such as to allow the outer layer **610** to be formed tightly onto the main inner structural layer **608**.

(41) In an embodiment, the main tubular shaft **102** can include an inner layer **612**. The inner layer **612** can include a polymer. The inner layer **612** can jacket or cover at least a portion of the inner portion of the main inner structural layer **608**. In an embodiment, the inner layer **612** can be at least 0.001 inches thick and not more than 0.005 inches thick. In an embodiment, the inner layer **612** can be at least 0.0007 inches thick. In an embodiment, the inner layer **612** can be at least 0.0005 inches thick. In an embodiment, the inner layer **612** can be no more than 0.007 inches thick. In an embodiment, the inner layer can be no more than 0.01 inches thick.

(42) In an embodiment, the inner layer **612** can include PTFE. In an embodiment, the inner layer **612** can include nylon. In an embodiment, the inner layer **612** can be coated with a hydrophilic polymer.

(43) In an embodiment, the outer layer **610** and the inner layer **612** can be fused together, such as through the main inner structural layer **608** and/or the braid **814** (shown in FIG. **8**).

(44) FIG. **8** shows a cross-section view of a portion of the main tubular shaft **802**, according to an embodiment. FIG. **9** shows a cross-section view from the end of the main tubular shaft **802**.

(45) In various embodiments, the main tubular shaft **802** can include a braided wire layer **814**. In an embodiment, the braided wire layer **814** can be disposed between the main inner structural layer **808** and the outer layer **810**. In an embodiment, the braided wire layer **814** can be disposed within a portion of the outer layer **810**. In another embodiment, the braided wire layer **814** is provided within the main inner structural layer **808** (e.g., along the interior of the layer **808**). Optionally, the braided wire layer **814** is between the inner layer **812** and the main inner structural layer **808**.

(46) The braided wire layer **814** can cover at least a portion of the main inner structural layer **808**, such as the helically wound multi-filar layer. In an embodiment, the braided wire can include stainless steel.

(47) In an embodiment, the braided wire layer **814** can be at least 0.0005 inches thick and not more than 0.010 inches thick. In an embodiment, the braided wire layer **814** can be at least 0.005 inches thick and not more than 0.010 inches thick.

(48) In an embodiment, the braided wire layer **814** can be at least 0.0004 inches thick. In an embodiment, the braided wire layer **814** can be at least 0.0003 inches thick. In an embodiment, the braided wire layer **814** can be no more than 0.015 inches thick. In an embodiment, the braided wire layer **814** can be no more than 0.020 inches thick.

(49) FIG. **10** shows a front view of a guide catheter **1000**, according to an embodiment. In an embodiment, the guide catheter **1000** can include a distal end curve **1016**. The distal end curve **1016** can be configured for anatomical conformance. The distal end curve **1016** can be heat processed in the metal portion of the catheter **1000**. In an embodiment, the main helically wound layer can terminate distally before a primary curve (e.g., one or more of curves **1018**, **1020**, **1022** or the like) of the distal end **112**.

(50) In various embodiments, the guide catheter can include a pre-shaped curve, such as a curved distal end region. The guide catheter can attain the pre-shaped curve configuration by heat-setting the metal in this portion of the catheter. The result can include a curve that retains its shape in body temperature and over time does not substantially soften, such as soften enough to unintentionally change shape.

(51) It should be noted that, as used in this specification and the appended claims, the singular forms “a,” “an,” and “the” include plural referents unless the content clearly dictates otherwise. Thus, for example, reference to a composition containing “a compound” includes a mixture of two or more compounds. It should also be noted that the term “or” is generally employed in its sense including “and/or” unless the content clearly dictates otherwise.

(52) It should also be noted that, as used in this specification and the appended claims, the phrase “configured” describes a system, apparatus, or other structure that is constructed or configured to perform a particular task or adopt a particular configuration to. The phrase “configured” can be used interchangeably with other similar phrases such as arranged and configured, constructed and arranged, constructed, manufactured and arranged, and the like.

(53) All publications and patent applications in this specification are indicative of the level of ordinary skill in the art to which this technology pertains. All publications and patent applications are herein incorporated by reference to the same extent as if each individual publication or patent application was specifically and individually indicated by reference.

(54) The technology has been described with reference to various specific and preferred embodiments and techniques. However, it should be understood that many variations and modifications may be made while remaining within the spirit and scope of the technology.

Claims

1. An interventional guide catheter for introducing interventional catheters into vasculature, comprising: a main tubular shaft extending between a proximal end and a distal end having a distal tip, the main tubular shaft including a catheter lumen configured for introducing catheters through the distal tip; wherein the main tubular shaft includes a sidewall having: an outer layer; an inner layer extending around the catheter lumen; a metal main inner structural layer including metal interposed between the outer and inner layer; and a braided wire layer surrounding the metal main inner structural layer, the braided wire layer within the outer layer; and wherein the outer layer, inner layer, metal main inner structural layer and the braided wire layer are a laminate, and the outer layer and inner layer are isolated from each other by layers consisting of one or more of the metal main inner structural layer or the braided wire layer.
2. The interventional guide catheter of claim 1, wherein the outer layer and inner layer isolated from each other includes the outer layer and inner layer are not fused.
3. The interventional guide catheter of claim 2, wherein the main inner structural layer includes a helically wound multi-filar wire.
4. The interventional guide catheter of claim 3, wherein the sidewall has a sidewall thickness including: an outer layer component thickness of around 0.001 to 0.005 inches; an inner layer component thickness of around 0.0005 to 0.0007 inches; and a main inner structural layer component thickness of around 0.0015 to 0.010 inches.
5. The interventional guide catheter of claim 4, wherein the braided wire layer has a braided wire layer thickness of around 0.0005 to 0.010 inches.
6. The interventional guide catheter of claim 5, wherein the braided wire layer thickness is around 0.005 to 0.010 inches.
7. The interventional guide catheter of claim 5, wherein the helically wound multi-filar wire includes distal and proximal edges, and the proximal edges of windings of the helically wound multi-filar wire are engaged with distal edges of proximate windings of the helically wound multi-filar wire.
8. The interventional guide catheter of claim 7, wherein the distal end includes a curve, and the main inner structural layer is absent from the curve of the distal end.
9. The interventional guide catheter of claim 7, wherein the distal end includes a curve, and the main inner structural layer is present in the curve of the distal end.
10. The interventional guide catheter of claim 7, wherein the helically wound multi-filar wire includes a swaged, round, square or rectangular cross-section.
11. The interventional guide catheter of claim 10, wherein a cross section of the main tubular shaft includes a cross-sectional metallic component of around 40 to 60 percent.
12. The interventional guide catheter of claim 11, wherein the braided wire layer includes a cross-

sectional metallic component of around 5 to 10 percent of the cross section of the main tubular shaft.
